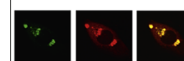


Available online at www.sciencedirect.com

ScienceDirect

www.elsevier.com/locate/brainres

Brain Research



Review

The role of visual deprivation and experience on the performance of sensory substitution devices



H. Christiaan Stronks^{a,b,*}, Amy C. Nau^c, Michael R. Ibbotson^{d,e}, Nick Barnes^{a,f}

^aComputer Vision Research Group, NICTA, Canberra, ACT 2601, Australia

^bDepartment of Neuroscience, The John Curtin School of Medical Research, Australian National University, Canberra, ACT 2601, Australia

^cKorb & Associates, 400 Commonwealth Ave., Unit 2, Boston, MA 02215, USA

^dNational Vision Research Institute, Australian College of Optometry, Melbourne, VIC 3053, Australia

^eDepartment of Optometry and Vision Sciences, University of Melbourne, Melbourne, VIC 3010, Australia

^fResearch School of Engineering, College of Engineering and Computer Science, Australian National University, Canberra, ACT 2601, Australia

ARTICLE INFO

Article history:

Accepted 23 June 2015

Available online 14 July 2015

Keywords:

Sensory substitution

Rehabilitation

Cross-modal plasticity

Training

Learning

Visual deprivation

ABSTRACT

It is commonly accepted that the blind can partially compensate for their loss of vision by developing enhanced abilities with their remaining senses. This visual compensation may be related to the fact that blind people rely on their other senses in everyday life. Many studies have indeed shown that experience plays an important role in visual compensation. Numerous neuroimaging studies have shown that the visual cortices of the blind are recruited by other functional brain areas and can become responsive to tactile or auditory input instead. These cross-modal plastic changes are more pronounced in the early blind compared to late blind individuals. The functional consequences of cross-modal plasticity on visual compensation in the blind are debated, as are the influences of various etiologies of vision loss (i.e., blindness acquired early or late in life). Distinguishing between the influences of experience and visual deprivation on compensation is especially relevant for rehabilitation of the blind with sensory substitution devices. The BrainPort artificial vision device and The vOICe are assistive devices for the blind that redirect visual information to another intact sensory system. Establishing how experience and different etiologies of vision loss affect the performance of these devices may help to improve existing rehabilitation strategies, formulate effective selection criteria and develop prognostic measures. In this review we will discuss studies that investigated the influence of training and visual deprivation on the performance of various sensory substitution approaches.

© 2015 Elsevier B.V. All rights reserved.

*Corresponding author at: NICTA, Tower A, Level 2, 7 London Circuit, Canberra, ACT 2601, Australia. Tel.: +61 2 6267 6334; fax: +61 2 6267 6220.

E-mail addresses: christiaan.stronks@nicta.com.au (H.C. Stronks), nauxac@gmail.com (A.C. Nau), mibbotson@nvri.org.au (M.R. Ibbotson), nick.barnes@nicta.com.au (N. Barnes).

<http://dx.doi.org/10.1016/j.brainres.2015.06.033>

0006-8993/© 2015 Elsevier B.V. All rights reserved.

Contents

1. Introduction	141
1.1. The effects of vision loss on the remaining senses	141
1.2. Sensory substitution	141
1.3. Purpose of the review	142
2. Effects of experience and blindness on sensory substitution device performance	143
2.1. Studies with the BrainPort artificial vision device	143
2.2. Studies using vibrotactile devices	147
2.3. Studies using Braille-like stimuli	147
2.4. Studies with visual-to-auditory substitution approaches	148
3. Discussion	148
Acknowledgments	150
Appendix A. Supplementary information	150
References	150

1. Introduction

1.1. The effects of vision loss on the remaining senses

Blind people are commonly believed to have superior perceptual skills in their remaining senses (Occelli et al., 2013; Rauschecker, 1995). A growing body of scientific evidence supports this notion (Leclerc et al., 2000; Occelli et al., 2013; Théoret et al., 2004), as the blind have been shown to have lower tactile thresholds (Sterr et al., 1998), better tactile acuity (Goldreich and Kanics, 2003; Van Boven et al., 2000), and improved auditory skills (Fieger et al., 2006; Lessard et al., 1998; Röder et al., 1999). These enhanced perceptual skills have been explained by cross-modal compensatory adaptive changes in the remaining intact sensory systems (Pascual-Leone et al., 2005; Rauschecker, 1995).

Blindness and the resulting visual deprivation of the brain results in dramatic changes in cortical topography (Bavelier and Neville, 2002; Kupers and Ptito, 2014; Noppeney, 2007; Pascual-Leone et al., 2005; Théoret et al., 2004) and brain regions associated with the remaining senses have been shown to expand into the deafferented visual cortices (Büchel et al., 1998; Leclerc et al., 2000; Melzer et al., 2001; Ptito et al., 2008; Sadato et al., 1998; Shimojo and Shams, 2001). This phenomenon allows the visual cortices to be activated by tactile or auditory input instead (Poirier et al., 2007b; Reich et al., 2012; Renier and de Volder, 2005), which can subsequently fulfill essential tasks in the processing of information from the remaining senses (Cohen et al., 1997; Hamilton et al., 2000). An underlying theory for cross-modal plasticity is that visual deprivation results in a loss of integrity of the deafferented pathways, a process that increases with duration of blindness (Lee et al., 2014). Indeed, neuroplastic changes have been shown to be dependent on the etiology of vision loss (i.e., blindness acquired early or late in life), and cross-modal activation of the visual cortices is more pronounced in early blind subjects compared to late blind individuals (Burton et al., 2002). For example, the primary visual cortex may be recruited in early blind subjects, but not in the late blind (Cohen et al., 1999; Sadato et al., 2002; Veraart et al., 1990).

One of the leading hypotheses on perceptual compensation suggests that the enhanced acuity of the remaining senses, at least in part, results from cortical plasticity

(Chebat et al., 2011; Cohen et al., 1997; Collignon et al., 2009; Maidenbaum et al., 2014; Van Boven et al., 2000; Wong et al., 2011). Indeed, expansion of sensory cortical areas may be accompanied by enhanced sensory acuity (Merzenich et al., 1983, 1984). However, the functional correlate of cortical reorganization following vision loss is a subject of debate (Lazzouni and Lepore, 2014; Röder et al., 1996) and cross-modal engagement of visual cortical areas in the blind may not necessarily be associated with enhanced performance at the functional level (Lee et al., 2014; Matteau et al., 2010; Ptito et al., 2005, 2009, 2012). These observations call into question the functional role of cross-modal neuroplastic changes in the visually deprived brain.

Compensation in the blind is mediated through experience (Maidenbaum et al., 2014; Sathian, 2000), as evidenced by studies performed in Braille-reading blind subjects. The tactile acuity of Braille-proficient blind subjects has been shown to be better than normally sighted controls in tasks where stimuli similar to Braille were used. However, when meaningless stimuli unrelated to Braille were used, acuities were not different between the blind and sighted (Alary et al., 2009; Grant et al., 2000; Heller, 1989). Conversely, normally sighted controls performed as well as the blind on an acuity task using Braille-like dot patterns after having received sufficient training (Grant et al., 2000). Moreover, blind Braille readers were shown to outperform normally sighted subjects in spatial acuity tasks on the fingers, but not on the lips (Wong et al., 2011). These results show that compensation may be experience-related, and that experience effects can be specific, and task-dependent (Sathian and Stilla, 2010). Hence, perceptual compensation in the remaining sensory systems may be explained by the fact that the blind rely more heavily on these remaining senses in everyday life.

1.2. Sensory substitution

Traditional methods aimed at improving the quality of life of blind people have typically relied on re-directing visual information to the remaining senses. Notable examples include the long cane and the Braille writing system, which are considered to be among the most successful rehabilitation strategies for the blind developed so far (Bach-y-Rita and Kercel, 2003). Both approaches are based on visual sensory

substitution, which refers to the process of obtaining information about the world through an alternative intact sensory channel (i.e., tactile sense) that would normally be obtained by the visual system (Bach-y-Rita, 1983).

More recently, artificial vision technologies based on sensory substitution have been developed to increase the amount of transmitted information. An early approach was the Tactile Vision Substitution System or TVSS (Fig. 1), which reconstructed camera images into binary tactile representations by means of an array of vibrotactile solenoid stimulators (Bach-y-Rita et al., 1969a, 2003). A similar concept is used in the BrainPort V100 artificial vision device (Danilov and Tyler, 2005), which features an array of electrodes that is placed on the tongue (Fig. 2). The BrainPort has been tested in a prospective, multi-center FDA safety study (<http://clinicaltrials.gov/ct2/show/NCT01488786>), and has received FDA market approval in the US, and it has received the CE Mark in Europe where it is commercially available.

The Optacon device is a vibrotactile reading machine for the blind developed by the Telesensory Corporation (Goldish and Taylor, 1974; Linvill and Bliss, 1966). The user deploys a camera to scan printed text, which is translated into tactile stimuli on the fingertip (Fig. 3). Remarkably high reading speeds of 70–100 words per minute have been reported with the device (Craig, 1977, 1981). Despite its apparent success, the production of the device was discontinued in 2000.

Several auditory substitution devices have been developed, and notable examples include The vOICe and the

prosthesis-substituting-vision-by-audition (PSVA) device. The vOICe translates images into sounds by re-mapping their x-axis into the time domain, and their y-axis into the frequency domain (Meijer, 1992). This “image-sweep line” technique has subsequently been deployed in the EyeMusic device for use in sensory substitution (Abboud et al., 2014). The PSVA makes use of a pixel-by-pixel acoustic frequency transformation, where vertical position is coded as pitch and horizontal position as binaural intensity and phase differences, and brightness into loudness (Capelle et al., 1998).

1.3. Purpose of the review

In this review we focus on the impact of visual deprivation and experience on visual compensation in the blind. We review specifically those studies that investigated training effects on the performance of various vision substitution approaches, and that compared these training effects between blind and sighted subjects, or between subjects with varying etiologies of blindness. In contrast to established sensory substitution approaches such as Braille, many people are naïve to novel sensory substitution devices like the BrainPort. Therefore, such devices offer an excellent opportunity to distinguish between experience effects and the impact of visual deprivation, since the influence of

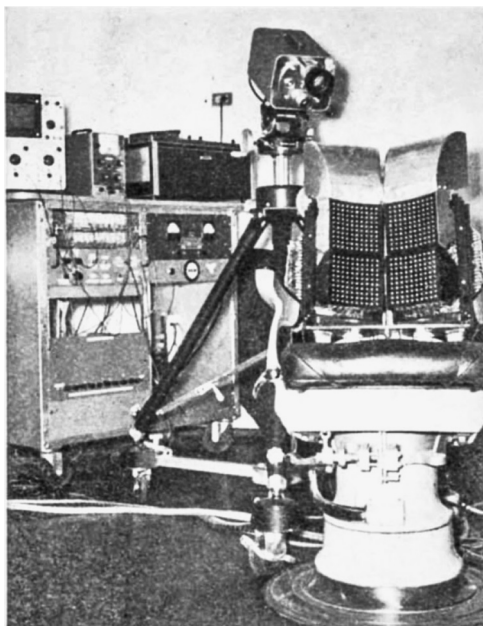


Fig. 1 – Original photograph of the Tactile vision substitution system (TVSS) developed by Bach-y-Rita and colleagues. It shows a television camera on a tripod behind the chair, control electronics on the left, and the tactile display consisting of 400 vibrotactile solenoid stimulators (tactors) mounted in the back of a dental chair. The device converted camera images into two-dimensional, binary vibrotactile representations presented onto the back of a subject. Adapted from Bach-y-Rita et al. (1969b) with permission from Macmillan Publishers, Ltd.

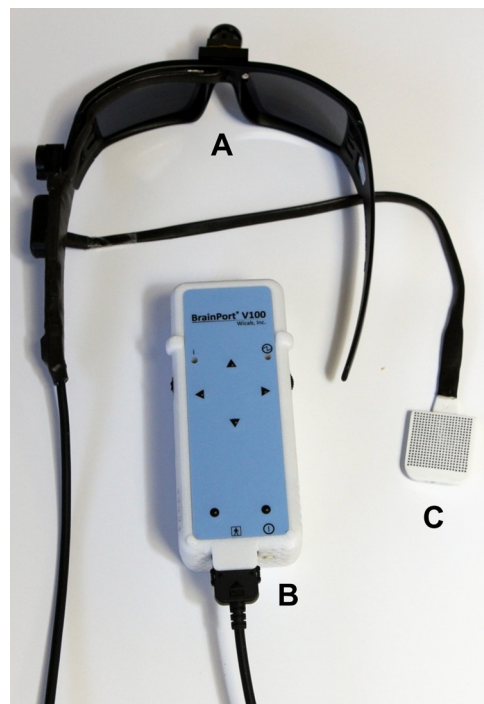


Fig. 2 – The BrainPort® V100 artificial vision device developed by Wicab. (A) Glasses-mounted camera, (B) control unit, and (C) intraoral display (IOD, previously called the tongue-display-unit or TDU). The IOD receives visual input through the glasses-mounted camera and vision processor. Advantages of the electrotactile approach are the fact that the tongue has excellent tactile resolution, and its low impedance facilitates low power consumption (Bach-y-Rita et al., 1998). Reprinted with permission from Wicab, Inc., Middleton, WI, USA. © 2013, Ken Sweitzer.

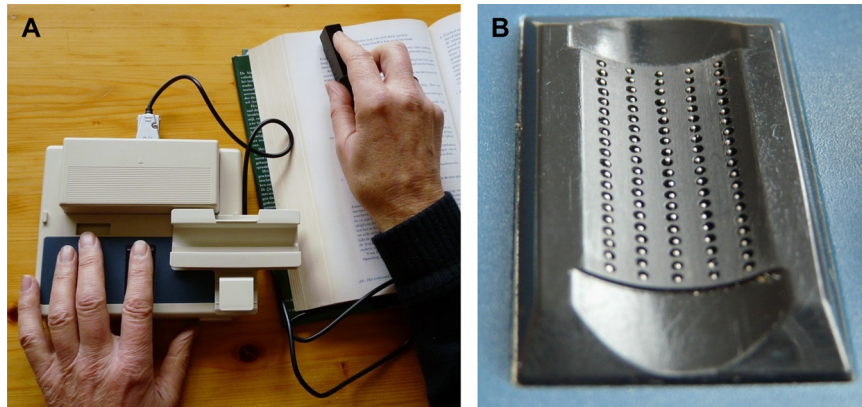


Fig. 3 – The Optacon II device developed by Telesensory Systems, Inc. (A) overview, (B) close up of the tactile display. A camera was used to scan regular printed text, which was translated into tactile representations on a 20×5 tactile array. The Optacon I featured a tactile display with 24×6 tactors. The Optacon was discontinued in 2000, and Telesensory Systems ceased its operations in 2005. © 2014, Stichting Nationaal Blindenmuseum (Foundation National Museum of the Blind, the Netherlands).

practice effects can be closely controlled. The extent to which experience and vision loss influence the performance of various sensory substitution approaches is important for improving rehabilitation strategies, devising of effective selection criteria and the development of prognostic measures for these devices. In addition, it is important to know whether the etiology of vision loss influences the ability to successfully use sensory substitution devices.

2. Effects of experience and blindness on sensory substitution device performance

In [Table 1](#) we have summarized the twenty vision-substitution studies we found that investigated the effects of training on sensory substitution device performance. Nineteen reports compared the effects of training between blind and sighted subjects, or between blind people with different etiologies of vision loss. Blind subjects in these nineteen studies were all reported to have residual bare light perception or less, i.e., they had no functional vision. One study used blindfolded normally sighted subjects only. The studies in [Table 1](#) are categorized according to device or approach, and grouped with respect to approximate task difficulty. The superior performing subject groups, if any, are indicated with shaded cells in the table.

Throughout this review we have used the term ‘early blind’ for both the congenitally blind (blind at birth) and those that became blind early in life. We grouped these two categories together, because the definition of congenital blindness differs across studies. Specifically, it is often used not only for subjects blind at birth, but also for those with loss of vision early in life (e.g., [Matteau et al., 2010](#); [Ptito et al., 2012](#); [Wan et al., 2010](#)). Early blindness was defined as blindness diagnosed at, or before the age of 14 years. Late blindness was defined by a diagnosis received after the age of fourteen ([Ortiz et al., 2011](#); [Wan et al., 2010](#)). These definitions acknowledge the critical period during the first 14 to 16 years of life in which cross-modal plasticity has been shown to

occur in the primary visual cortex ([Cohen et al., 1999](#); [Sadato et al., 2002](#)).

Out of the nineteen studies that included blind subjects, fourteen explicitly included only subjects with vision loss due to peripheral origins ([Table 1](#)). Only one study makes specific mention of including a subject who’s blindness was caused by a central disorder, namely a cerebral hematoma ([Chebat et al., 2007](#)). The remaining four studies included one or more subjects with unspecified etiologies ([Chebat et al., 2011](#); [Ortiz et al., 2011](#); [Renier and De Volder, 2010](#); [Sampaio et al., 2001](#)). While the great majority of the subjects covered in this review suffered from blindness of peripheral origin ([Table 1](#)), specific etiologies varied considerably between and within studies. An overview of the most commonly encountered etiologies can be found in [Table 2](#) in the supplementary information.

2.1. Studies with the BrainPort artificial vision device

Rehabilitation with the BrainPort device has been shown to depend heavily on appropriate training strategies ([Nau et al., 2015](#)). Anecdotal findings with the predecessor of the BrainPort, the TVSS ([Bach-y-Rita et al., 1969a](#)) have provided early evidence for the necessity of appropriate training. The TVSS enabled blind subjects to perform a variety of tasks normally associated with vision, such as recognizing objects and faces, and enabled blind subjects to appreciate abstract concepts such as depth, zooming and parallax ([Bach-y-Rita, 1979, 1983](#); [Bach-y-Rita et al., 2003](#); [Lenay et al., 2003](#)). However, to obtain this variety of capabilities, substantial practice was necessary and training times of 20 up to 150 h have been reported with the TVSS ([Bach-y-Rita et al., 1969a](#); [Bach-y-Rita, 1983](#)). Other groups that have developed general-purpose rehabilitation strategies have reported comparably long training times: 30 h for a pneumatic variant of the TVSS ([Segond et al., 2013](#)), up to 60 h for the BrainPort device ([Nau et al., 2015](#)), and 33 to 101 h for The vOICe ([Striem-Amit et al., 2012](#)). A recent study shows that daily use of the BrainPort may improve the performance with the device over the course of 3–6 months ([Nau et al., 2015](#)).

Table 1 – List of the 20 reviewed studies that investigated effects of training on sensory substitution device performance. Studies are sorted on sensory substitution approach and task. Shaded cells indicate a superior performance before (pre) or after training (post).

#	Device/ approach	Task	Onset of blindness (n)	Etiology (n)	blind (n)	NS (n)	Training (h)	Superior group		Study
								Pre	Post	
1	TDU	Acuity	EB (6)	Unspecified (6) ^A	6	6	9	–	–	Sampaio et al. (2001)
2	BrainPort	Acuity	EB (6)	Peripheral (6)	6	5	7	–	–	Ptito et al. (2005)
3	Brainport	Acuity	EB (15)	Peripheral (14) Central (1)	15	25	0.2	n.t.	–	Chebat et al. (2007)
4	BrainPort	DOM	EB (8) LB (5)	Peripheral (13)	13	8	2.5–7 ^B	–	–	Kupers et al. (2006)
5	BrainPort	DOM	EB (7)	Peripheral (7)	7	6	7	–	–	Ptito et al. (2009)
6	BrainPort	DOM	EB (8)	Peripheral (8)	8	9	2.5	–	–	Matteau et al. (2010)
7	BrainPort	Various	EB (3) LB (21)	Peripheral (24)	24	6	15 ^G	–	–	Nau et al. (2013)
8	BrainPort	Various	EB (11) LB (34)	Peripheral (46)	46	6	15 ^G	–	–	Lee et al. (2014)
9	BrainPort	Shape recognition	EB (8)	Peripheral (8)	8	10	5	–	–	Ptito et al. (2012)
10	BrainPort	Navigation	EB (10)	Peripheral (10)	10	10	? (60 trials)	–	EB	Kupers et al. (2010)
11	BrainPort	O&M	EB (16)	Peripheral (12) Unspecified (4) ^A	16	11	Short familiarization	n.t.	EB	Chebat et al. (2011)
12	BrainPort	O&M	EB (3) LB (27) ^C	Peripheral (36)	36	6	15 ^G	B	–	Nau et al., 2014
13	Vibro-tactile	Line orientation	EB (9) LB (9)	Peripheral (17) Unspecified (1) ^A	18	10	65	–	–	Ortiz et al. (2011)
14	Pneumatic	Depth perception	EB (6)	Peripheral (6)	6	7	30 ^G	NS	–	Segond et al. (2013)
15	Braille-related	Masking	EB (33) LB (24)	Peripheral (57)	57	89	Braille ^D	n.t.	EB	Bhattacharjee et al. (2010)
16	Braille-related	Frequency discrimination	EB (22) LB (11)	Peripheral (33)	33	33	Braille ^D + training (unspecified)	n.t.	EB ^E	Wan et al. (2010)
17	Braille	Braille recognition	–	–	–	24	≥ 30	–	NS	Kauffman et al. (2002)
18	Echo location	Location	EB (5)	Peripheral (5)	5	4	6	n.t.	–	De Volder et al. (1999)
19	PSVA	Pattern recognition	EB (6)	Peripheral (6)	6	6	12	–	EB	Arno et al. (2001a, 2001b) ^F
20	PSVA	Depth perception	EB (10)	Peripheral (8) Unspecified (2) ^A	10	20	~4	NS	–	Renier and De Volder (2010)

EB: early blind (≤ 14 years).

LB: late blind (> 14 years).

B: blind (EB and LB).

NS: normally sighted.

DOM: direction of motion.

O&M: orientation and mobility.

PSVA: prosthesis-substituting-vision-by-audition device.

TDU: tongue display unit (earlier version of the intra-oral display used in the BrainPort).

n.t.: not tested.

^A These studies did not specify whether the pathology was peripheral or central, e.g. medical accidents (Chebat et al., 2011) and optic pathway bilateral lesion (Ortiz et al., 2011).

^B Training duration was not specified and estimated based on other publications where the same DOM task was used (Matteau et al., 2010; Ptito et al., 2009).

^C 24 Subjects in Nau et al., 2014 were identical to those in Nau et al. (2013). The onsets of blindness in 6 out 12 additional subjects were provided for this review by A. C. Nau and J. Fisher. The other 6 had unknown onsets of blindness.

^D Bhattacharjee et al. (2010) and Wan et al. (2010) compared Braille-proficient blind subjects with sighted subjects not proficient in Braille.

^E Subgroup with very early onset of blindness < 0.5 years (Wan et al., 2010).

^F Arno et al., 2001a, 2001b's two studies are counted as one, because the tasks and study subjects were identical.

^G General rehabilitation regime (as opposed to task-oriented training used in other studies).

The effects of training and blindness on visuo-tactile performance have been extensively addressed with the BrainPort. A number of early studies deploying relatively small study populations addressed these issues using pattern recognition tasks. One of these studies compared six early blind subjects with five normally sighted, blindfolded controls using a tongue display unit (TDU) comparable to the intraoral display (IOD) of the BrainPort (Ptito et al., 2005). The subjects were tested on a 4-alternative forced choice (4AFC) tumbling 'T' task before, during and after 7 h of task-specific training. Correct rates increased from around chance level to ~90% in both groups, while reaction times decreased from ~50 to ~20 s in the control group, and from ~40 to ~5 s in the blind group (Fig. 4). Differences between the learning curves in both groups were statistically insignificant.

In a similar study (Ptito et al., 2012), shape recognition scores were determined using a 4AFC task in a blind group (8 early blind) and a sighted control group ($n=10$) before and after 150 min of training (Fig. 5). Recognition scores were shown to increase from ~55% to ~90% in both groups, while recognition times decreased from 10–15 s to 5–10 s. Learning curves did not differ significantly between the blind and sighted groups.

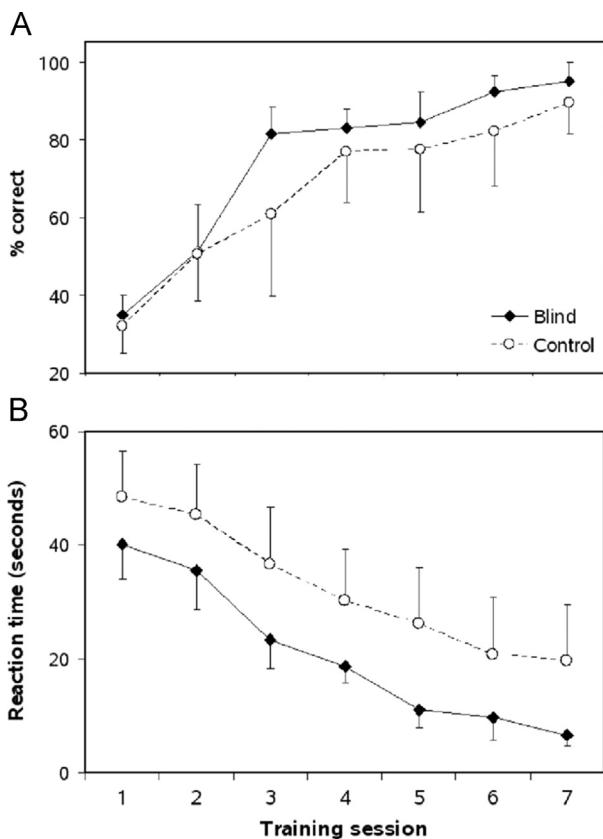


Fig. 4 – Learning curves for an orientation discrimination (tumbling 'T') task in early blind and blindfolded controls using a four-alternative, forced choice paradigm. (A) Correct rates. (B) Response time. No significant differences in performance were observed between the groups. Graphs reproduced from Ptito et al. (2005) by permission of Oxford University Press and the authors.

A set of closely related studies with the tongue display unit deployed a 2AFC direction-of-motion task, and compared training effects between blind and sighted subjects. In one of these studies, correct rates increased from ~55% to ~90% in both the early blind ($n=8$) and the sighted ($n=9$) after having received 150 min of practice (Matteau et al., 2010). Using a longer training period of 7 h (Ptito et al., 2009), it was shown that correct rates reached 95% in early blind ($n=7$) and 90% for sighted controls ($n=6$). Using the same task in eight early blind, five late blind and eight sighted subjects, the authors showed that all but three subjects (2 blind, 1 sighted control) reached at least 85% correct rates after a week of training, while before training performance was at chance level (i.e., 50%). The authors make no mention of training time in this report (Kupers et al., 2006). In these three studies no significant differences were found between the performance of the blind (mostly early blind) and sighted controls.

Two studies have estimated visuo-tactile acuities with the tongue display unit. One such study compared Snellen Tumbling 'E' acuities in 25 controls and 15 early blind subjects after a short 10-min familiarization period with line and optotype stimuli (Chebat et al., 2007). The results showed an insignificant difference between the acuities in the blind group (1/206, or 20/4100) compared to the controls (1/237 or 20/4700). The slightly better performance in the blind group could be explained by the fact that a subset of the best performing blind patients had prior experience with the BrainPort device on a similar task.

An earlier report deploying a Snellen tumbling 'E' visuo-tactile acuity task with the tongue display unit (Sampaio et al., 2001) compared six early blind subjects and six normally sighted, blindfolded subjects, all naïve to the system. The acuity scores of the blind group were slightly better, but did not differ significantly from the control group before training, as the average visuo-tactile acuity was 20/860 in both groups. One average-performing subject from each group was then trained for 9 h using a line detection and line orientation task, which doubled the acuity to an average of 20/430 in both subjects.

A recent study by Nau et al. (2013) evaluated the effects of 15 h of training with the BrainPort device on 11 computer-based psychophysical tests including 24 blind subjects and 6 normally sighted controls. The training regimen was a general rehabilitation protocol for BrainPort users and was not specifically targeted to improve the performance with the specific test stimuli. Before training the subjects performed typically at chance level. After training, their performance significantly increased on 4 out of 11 tests, namely light perception (50 to 92%), location (13 to 46%), grating acuity (25 to 77%), and visual acuity determined with a white tumbling 'E' on a black background (2.60 to 2.57 logMAR, or 20/8000 to 20/7400 Snellen equivalent). Time resolution and motion task performance increased (from 50 to 56% and from 13 to 17%, respectively), but not significantly. Visual acuity on the Landolt C test (black-on-white and vice versa) or tumbling 'E' (white-on-black) did not improve, nor did the contrast acuity of a white 'C' on a black background, and vice versa. The authors explain the relatively poor results on the time resolution and motion tasks based on the limited temporal resolution of the BrainPort. The lack of learning effects on the Landolt C test was explained by the limited spatial resolution of the intra-oral display. Comparison of the performance

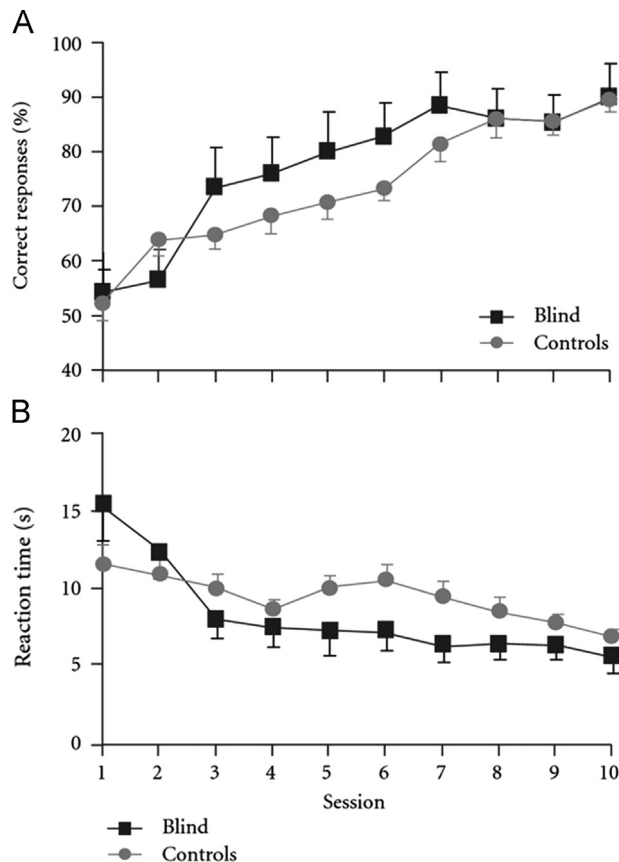


Fig. 5 – Learning curves for a shape recognition task in early blind and blindfolded controls using a four-alternative forced choice paradigm. The stimuli were a square, rectangle, triangle, and an ‘E’. (A) Mean percentage changes $\pm$ standard error of the mean (SEM) of correct responses and (B) mean reaction times $\pm$ SEM. No significant differences in performance were observed between the groups. Graphs reproduced from [Ptito et al. \(2012\)](#) under a Creative Commons Attribution License, and with permission of the authors.

between subjects blind for 10 years and those blind for less than 10 years (including the sighted controls) showed no significant difference in any of the 11 tests.

A subsequent study by the same group with 52 subjects (46 blind, 6 sighted controls) and a subset of six tests from the original test battery showed that the 15-h training regime with the BrainPort significantly increased light perception (from 54 to 83%), temporal resolution (51 to 64%), location estimation (12 to 43%), grating acuity (25 to 64%) and visual acuity using a tumbling ‘E’ (2.70 to 2.49 logMAR, or 20/10,000 to 20/6200). Motion detection did improve (from 13 to 16%), but not significantly ([Lee et al., 2014](#)). In line with their 2013 study, the authors did not find significant correlations between task performance and onset (or duration) of blindness, or between task performance and etiology of blindness on any of the six tests. This lack of correlation was found both before and after training.

The BrainPort has also been shown to be useful in more complex tasks, such as object localization, depth perception and navigational tasks ([Arnoldussen and Rhode, 2010](#);

[Arnoldussen et al., 2012](#); [Segond et al., 2005](#)). Navigation and orientation-and-mobility with the BrainPort and related devices are, however, limited by poor resolution and few studies have made an attempt to investigate the effects of experience and blindness on these tasks. In one study the performance of an early blind group ($n=16$) and normally sighted controls ($n=11$) were compared in an obstacle-detection and avoidance task. It was shown that the blind outperformed the blindfolded controls in terms of detection and avoidance efficiency ([Chebat et al., 2011](#)). The study did not include a formal training session and subjects only received a short familiarization period in which they performed cognitively less demanding acuity tasks. The subjects were only tested after device familiarization.

In another study, deploying a virtual navigation paradigm, a formal training period was used ([Kupers et al., 2010](#)). Ten early blind subjects and ten normally sighted controls, all naïve to the task, were instructed to learn two closely related navigational tasks over the course of 4 days (15 trials per test, per day). One task involved re-drawing of a simple 2-corner route, while in the other task they had to distinguish the learned route from another. Overall performance was not statistically different between the blind and the sighted, and correct rates increased from 60% to 80–90% in both groups for the re-drawing task, and from 20% to 100% for the discrimination task. On the last day of training the blind performed significantly better on both tests (90 vs 80%, and 100 vs 90%). However, when formally tested after training (day 5) on the discrimination task, both groups scored equally well (i.e. at around 100%). The re-drawing task was not tested. Hence, we conclude that the superiority of the early blind was marginal. Since the study does not provide training times (apart from the number of trials) it is difficult to compare the amount of training with related studies.

A recent study by [Nau et al. \(2014\)](#) investigated the impact of training on an orientation-and-mobility task using an obstacle course in 36 blind subjects and 6 normally sighted controls. The training consisted of 15 h of general rehabilitation that was not specifically designed for orientation-and-mobility. After training, orientation-and-mobility skills improved significantly, as indicated by increased preferred walking speeds and fewer collisions. Preferred walking speeds did not significantly differ between the sighted and the blind before or after training. Collision rates were initially lower in the blind, but were not significantly different after training, indicating that the blind may have been advantaged before training in the orientation-and-mobility task due to experience effects from everyday life.

We conclude that from the 12 studies that have examined the BrainPort, only two showed a functional difference between the blind and the sighted. One of these ([Kupers et al., 2010](#)) revealed a superiority in the early blind over the sighted. However, this advantage was diminished on the last day of the experiment for one of two closely related tasks, given that the performance of both groups was nearly perfect. The other task was not tested on the last day. We argue that the learning phase may have been too short to reach its full effect, which could have placed the blind in an advantageous position, since they had been relying on alternative sources of information in everyday life. The other study by [Chebat et al.](#)

(2011) did not include a formal training session which may have placed the early blind group in a similar advantageous position.

2.2. Studies using vibrotactile devices

Substantial experience effects have been shown for the Optacon device and other vibrotactile approaches (Apkarian-Stielau and Loomis, 1975; Cholewiak and Craig, 1984; Cholewiak and Collins, 1997; Epstein et al., 1989), but we are aware of only two studies that compared experience effects in the blind and sighted. One of these deployed a training period of 3 months with 1-h sessions for 5 days per week (i.e., approximately 65 h of training) on the performance with a vibrotactile device held on the palm of the hand. The 18 blind subjects had various etiologies (early and late blind) and were compared with 10 sighted controls. Both groups were trained and tested on a 2AFC line-orientation discrimination task (Ortiz et al., 2011). After the training period, correct rates on the task improved from 78% to 98% in the blind, and from 83% to 98% in the controls. No significant difference was found between the sighted group and the blind population. No attempt was made to stratify the blind group by onset of blindness.

The other study made use of a pneumatic variant of the TVSS and compared 6 early blind subjects with 7 sighted controls. Both groups received 30 h of general-purpose rehabilitation training and were trained and tested on several tasks during the training. Both groups showed equal performance and comparable training effects in a pattern recognition task. On a zooming and perspective task the early blind had more difficulties with monocular depth cues before training. After training, however, both groups performed equally well. Navigation task performance was similar between both groups and increased similarly after training (Segond et al., 2013).

2.3. Studies using Braille-like stimuli

Many studies deploying acuity tasks related to Braille reading have found that blind subjects proficient in Braille outperformed normally sighted subjects not proficient in Braille. Consequently, many of these studies have concluded that the superiority of the blind was, at least in part, due to Braille-reading experience (Alary et al., 2009; Grant et al., 2000; Heller, 1989; Sterr et al., 1998; Wong et al., 2011). We found three studies using Braille-related tasks that investigated the effects of visual deprivation whilst controlling for experience effects.

In one study an attempt was made to distinguish between the effect of experience and visual deprivation by using tactile stimuli delivered to the fingertips that resembled Braille stimuli in the temporal domain, but not in the spatial domain (Bhattacharjee et al., 2010). The study included 57 blind subjects with variable proficiency in Braille reading, and a group of 89 normally sighted controls. When both groups were tested on a tactile detection and amplitude discrimination task no differences were found. In contrast, when they were compared on their temporal capabilities using a forward-masked detection task the blind were shown to significantly outperform the sighted. Moreover, the performance on this task was positively

correlated with Braille reading proficiency. The different outcomes on the spatial and temporal tasks provide evidence that the blind outperformed the sighted in Braille reading through experience effects. To investigate the effect of etiology of blindness, the authors divided the group of blind subjects according to onset of blindness. None of the groups differed from the sighted controls in the spatial domain. In the masking task it was shown that the early blind outperformed the sighted, while the late blind and non-Braille readers did not. Based on these results, the authors concluded that early onset blindness promotes accelerated tactile processing, in turn enhancing Braille reading speed. However, the early blind group also contained the fastest Braille readers, obfuscating their conclusion that this group was advantaged due to their etiology. To strengthen their argument that it was onset-of-blindness rather than experience, the authors showed evidence that masking effects did not differ between preferred and non-preferred Braille-reading fingers and hence that experience was not an important factor. However, the validity of this argument is contentious, since training-related performance improvements may transfer between digits (Sathian and Zangaladze, 1997) and even across hands (Spengler et al., 1997), which the authors discuss in their paper (Bhattacharjee et al., 2010).

Another attempt to address the interaction between onset-of-blindness and Braille proficiency was made in a study deploying a 2AFC vibrotactile frequency discrimination task unrelated to Braille stimuli. The stimuli were delivered simultaneously to three fingers on both hands (Wan et al., 2010). The study included a group of 33 early and late blind subjects with varying Braille proficiencies, and age and gender-matched normally sighted controls. All subjects were naïve to the task and a training session of undefined duration on the task was performed prior to testing. The early blind were shown to outperform the sighted controls, while late-blind individuals performed equally well as the sighted. There was a significant positive relationship between the time spent daily on Braille reading and performance in the blind cohort, indicating that active engagement in Braille reading favored better performance. Total duration of active Braille reading was, however, not correlated, showing that it was current Braille reading activity rather than long-term experience that determined superior tactile performance. After correction of daily Braille reading times it was shown that a subpopulation of early blind people (onset < 0.5 years, correct rate 83%) outperformed the early blind with later onset (1.4–13 years; correct rate 74%) and the late blind (correct rate 73%). These results point toward the possibility that visual deprivation starting very early in life leads to enhanced tactile acuity.

The third study we found directly addressed the effect of visual deprivation on Braille training. In this study, the authors blindfolded 24 normally sighted subjects for 5 consecutive days and compared their performance in a Braille discrimination task with non-blindfolded, normally sighted subjects. Both subject populations were divided into two groups. One group did not receive training and the other received at least 30 h of Braille training. While the sighted untrained group remained at 40% error rates, the sighted trained group improved to 35%. The blindfolded group improved to 20–25%, regardless of training. Hence, the

authors conclude that blindfolding resulted in better performance and that the enhancement was independent of Braille training (Kauffman et al., 2002). These results seemingly support the notion that the lack of vision alone can drive the development of superior tactile skills. However, as the authors discuss in their report, both the trained and untrained blindfolded subjects relied on their tactile sense in activities of daily living, and hence could have additional experience effects regardless of Braille training.

We conclude that the results from Bhattacharjee et al. (2010) and Kauffman et al. (2002) were not sufficiently controlled for experience effects to allow decisive conclusions to be made on the influence of visual deprivation or etiology of vision loss on performance. The study from Wan et al. (2010) yielded convincing evidence that the early blind, and especially those with very early onset of blindness may be advantaged over the sighted, as well as the late and early blind.

2.4. Studies with visual-to-auditory substitution approaches

Studies with auditory vision-substitution approaches have unequivocally shown the importance of training in normally sighted subjects (Arno et al., 1999; Brown et al., 2011; Kim and Zatorre, 2008, 2010; Levy-Tzedek et al., 2012; Poirier et al., 2006a, 2007a, 2006b; Renier et al., 2005). We have found four studies that compared experience effects between blind subjects and sighted controls.

In one study an echolocation device was used in a localization task. The study included a group of five early blind subjects and four normally sighted subjects. All subjects received 6 h of training on the task (De Volder et al., 1999). The authors did not statistically compare the functional results between the two groups, but mention that both groups consistently had correct rates of more than 90%. Hence, the difference between the blind and the normally sighted, if any, was not substantial.

Two closely related studies with the PSVA system compared training effects in a group of early blind ($n=6$) and sighted controls ($n=6$) before and after 12 h of task-related training on a 25-pattern recognition task. Recognition scores increased from approximately 1–2% in both groups to approximately 69% in the sighted and 91% in the blind. The difference after training was statistically significant (Arno et al., 2001a, 2001b). The two studies describe data on the same task using the same subjects, and the two studies are counted as a single study in this review.

Another study used the PSVA on a depth-perception task and compared the performance of 10 early blind to that of 20 normally sighted controls. They used an object-localization task, and total experimental time (including test and practice time) was approximately 4 h. The authors showed that the early blind had significantly higher error rates (17%) compared to the sighted controls (10%) before training. After training, the early blind significantly improved their performance, while the sighted showed no change in performance. Post-training error rates were approximately 8% in both groups (Renier and De Volder, 2010).

3. Discussion

All but one of the twenty reviewed studies (i.e., the study of Kauffman et al., 2002) have confirmed the importance of training on performance with a sensory substitution device. The lack of training effects in the one isolated study is difficult to reconcile with the overwhelming evidence that exists to date for the importance of rehabilitation and practice in any sensory substitution approach. Indeed, the authors of this study report effects of “over-training”, as evidenced by a numbness of the reading fingers in the trained groups, which may well have obscured the effects of Braille training. Two additional studies recognized the absence of significant practice effects in a subset of tests in a psychophysical test battery. The authors, however, attributed the lack of training effects to overcomplicated tasks where performance was limited by hardware constraints (Lee et al., 2014; Nau et al., 2013).

With regard to the impact of visual deprivation we conclude that none of the twelve studies that used the BrainPort have yielded conclusive support for an advantage of the blind over the sighted. Although two studies report a statistically significant difference between the blind and the sighted (Chebat et al., 2011; Kupers et al., 2010), the study designs did not allow for robust control of experience effects. The two vibrotactile studies we found did not report any significant effects of visual deprivation on performance. Out of the three discussed Braille studies, only one study yielded solid evidence for a superior performance of the early blind (<0.5 years) over those that became blind later in life and the fully sighted (Wan et al., 2010). In the area of auditory substitution devices one out of three studies conclusively showed an advantage for the early blind (Arno et al., 2001a, 2001b). Note that the two Arno et al. studies with the PSVA are counted as one, as they describe a single data set. The PSVA uses various acoustic parameters to encode visual information, including binaural cues for encoding spatial information of an image (Capelle et al., 1998). Indeed, early blind subjects have been shown to be more sensitive to binaural sound localization cues than normally sighted controls (Simon et al., 2002), and to perform better on an auditory spatial attention task (Röder et al., 1999). The enhanced sensitivity to binaural acoustic cues may (partly) account for the superior performance of the early blind with the PSVA.

We conclude that from the twenty studies discussed, six (30%) showed evidence of visual deprivation being positively correlated with performance. Out of these, we argue that only two (10%) showed conclusive evidence of a superiority in the blind after taking experience effects into account (Arno et al., 2001a, 2001b; Wan et al., 2010). Interestingly, both studies indicate that it is the early blind that outperform the late blind or the sighted. Results from imaging studies have suggested that the first 14 to 16 years represent a critical period in life for a functional shift from visual to tactile processing (Cohen et al., 1999; Sadato et al., 2002). However, in the study of Wan et al. (2010) it was specifically the blind with an onset very early in life (<0.5 years) that showed greater tactile abilities. Notably, they outperformed a subgroup of early blind with onsets between 1.4 and 13 years.

Both groups had an age range well within the critical period as determined by imaging studies (Cohen et al., 1999; Sadato et al., 2002).

In the majority of the twenty studies examined, sighted controls performed as well as their blind counterparts after training in terms of functional outcomes, despite of a lack of visual cortical activation. Prospective investigations will be necessary to firmly establish the findings of superior tactile skills in the early blind over the late blind, while additional imaging studies may refine the critical period that may shed light on the findings of Wan et al. (2010) at the neural level.

Two studies revealed superior performance in the normally sighted (Renier and De Volder, 2010; Segond et al., 2013). Interestingly, both of these studies investigated depth perception. It has indeed been shown that the early blind have more difficulty with interpreting depth information (Renier and De Volder, 2010) and the spatial skills of blind people in general is thought to be compromised (Kolarik et al., 2013; Occelli et al., 2014; Thinus-Blanc and Gaunet, 1997). It has been proposed that visual experience in early life is essential for shaping the other senses by providing them with feed-forward visual input (Fine et al., 2003; Rauschecker, 1995), which may therefore be especially relevant to the early blind (Liu et al., 2007; Occelli et al., 2013; Pascual-Leone et al., 2005). Indeed, both studies that found superior performance in the normally sighted compared them to a cohort of early blind (Renier and De Volder, 2010; Segond et al., 2013). More importantly, however, both studies revealed that the blind performed as well as the sighted subjects after training, again emphasizing the importance of experience.

It must be noted that many of the studies included only relatively small subject populations. While the non-invasive nature of sensory substitution devices facilitate larger numbers of potential subjects compared to visual prosthetics, recruiting a sufficient number of completely blind subjects that can be matched by age, duration and etiology of blindness is very difficult. Indeed, many of the cited BrainPort studies were, in fact, primarily imaging studies and population size was likely chosen on the basis of imaging power typical for fMRI and other imaging techniques. Hence, a lack of statistical power may have played a role in the assessment of the functional performance of the blind and the sighted in these studies. In addition, variables such as age, cognitive capabilities, mobility, transportation barriers and the willingness to undergo intensive training may complicate comparisons between sighted and blind populations. Ideally, collaboration between investigators in distinct geographic locales in prospective, randomized multi-center clinical trials would be a valuable strategy to sort out the questions about the mechanisms and functional outcomes of cross-modal interactions which have yet to be answered by the current body of literature.

An important consideration for understanding the neurobehavioral consequences of the use of sensory substitution devices are the pathways through which the visual cortex is activated. In the case of retinal implant strategies (Stronks and Dagnelie, 2014), visual information is conveyed through the primary visual pathways to V1 (Eckhorn et al., 2006). Therefore, trans-neuronal degeneration that increases with duration of loss of sensory function is likely to have a profound effect on the success of these approaches. In cochlear implants it has

indeed been shown that the timing of implantation is an important consideration (Nadol, 1997; Sharma et al., 2002). However, in the case of sensory substitution the visual cortex is secondarily activated after initial processing (Sadato et al., 2002) through somatosensory or auditory cortices. Therefore, if the originating tactile or auditory processing is normal, the ability to interpret the sensory substitution stimulus should not be substantially affected by the status of the primary visual cortex. Indeed, normally sighted individuals are able to successfully interpret information from sensory substitution devices, even when there is relative de-activation of their visual cortical regions as measured by BOLD fMRI (Murphy et al., 2014, 2015). The fact that the visual cortex is essentially “disengaged” when using sensory substitution devices in the sighted, but that their functional outcomes are equal to those that are blind, reinforces the notion that the primary visual cortex is not essential for sensory substitution device use, at least for the outcomes in the current body of literature. There exist numerous bidirectional inputs between secondary and tertiary visual cortices and other brain centers that may not be influenced by vision loss, especially when it is acquired later in life. These results suggest that as long as visual association areas and the primary channels for receiving sensory substitution inputs are intact, the status of V1 may not be critical for predicting success with sensory substitution.

While acknowledging the mentioned heterogeneity and small sample size of some of the blind populations tested in the studies reviewed, we conclude that based on the currently available body of knowledge, blindness or etiology of vision loss have no robust quantifiable benefits on sensory substitution device performance. This conclusion holds for a variety of tasks including standardized psychophysical tests including BaLM, BaGA and FrACT (Nau et al., 2013), as well as for cognitively more demanding tasks such as pattern recognition (Ptito et al., 2012). Most notably, even tasks involving depth perception and analyzing complex scenes for orientation-and-mobility purposes have not been convincingly shown to be different between the early blind and the normally sighted (Chebat et al., 2011; Kupers et al., 2010; Nau et al., 2014). Therefore, we claim that visual deprivation and onset of blindness have little predictive value for performance when using sensory substitution devices. The two exceptions to this conclusion are an improved vibrotactile acuity in the fingertips and enhanced auditory pattern recognition in the early blind compared to the late blind or the sighted (Arno et al., 2001a, 2001b; Wan et al., 2010). As the technology and outcome measures mature, studies which explore whether sensory substitution devices are useful to improve the performance of activities of daily living will be undertaken. It will be interesting to determine whether the blind or sighted subgroups perform better with sensory substitution devices in these relatively more complex situations.

We have shown substantial evidence for the positive effects of training. This finding stresses the importance of appropriate rehabilitation strategies for sensory substitution devices, which are in fact just emerging (Nau et al., 2014, 2015). In addition, our findings emphasize that the prognosis of an individual to successfully use a sensory substitution device will depend more on his or her willingness and mental capabilities to go through an intensive rehabilitation process,

than on the duration or underlying etiology of vision loss. As a consequence, selection criteria for device candidacy should focus more on the individual's psychological characteristics, rather than on the etiology or duration of vision loss.

Acknowledgments

We wish to thank Daniel Goldreich and Jacqueline Fisher for making available the data on the disease etiologies in their study subjects. NICTA is funded by the Australian Government through the Department of Communications and the Australian Research Council through the ICT Centre of Excellence Program. NICTA is also funded and supported by the Australian Capital Territory, the New South Wales, Queensland and Victorian Governments, the Australian National University, the University of New South Wales, the University of Melbourne, the University of Queensland, the University of Sydney, Griffith University, Queensland University of Technology, Monash University and other university partners. Ibbotson is supported by the Centre of Excellence for Integrative Brain Function (CE140100007) and gratefully acknowledges the support of the Lions Clubs of Victoria.

Appendix A. Supplementary information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.brainres.2015.06.033>.

REFERENCES

- Abboud, S., et al., 2014. EyeMusic: introducing a 'visual' colorful experience for the blind using auditory sensory substitution. *Restor. Neurol. Neurosci.* 32, 247–257.
- Alary, F., et al., 2009. Tactile acuity in the blind: a closer look reveals superiority over the sighted in some but not all cutaneous tasks. *Neuropsychologia* 47, 2037–2043.
- Apkarian-Stielau, P., Loomis, J.M., 1975. A comparison of tactile and blurred visual form perception. *Percept. Psychophys.* 18, 362–368.
- Arno, P., et al., 1999. Auditory coding of visual patterns for the blind. *Perception* 28, 1013–1029.
- Arno, P., et al., 2001a. Occipital activation by pattern recognition in the early blind using auditory substitution for vision. *NeuroImage* 13, 632–645.
- Arno, P., et al., 2001b. Auditory substitution of vision: pattern recognition by the blind. *Appl. Cogn. Psychol.* 15, 509–519.
- Arnoldussen, A., Rhode, K., 2010. Enhanced perception with the BrainPort vision device. *Invest. Ophthalmol. Vis. Sci.* 51, 3634 ARVO E-Abstract.
- Arnoldussen, A., et al., 2012. Perception of depth using the BrainPort vision device. *Invest. Ophthalmol. Vis. Sci.* 53, 325 ARVO E-Abstract.
- Bach-y-Rita, P., et al., 1969a. Vision substitution by tactile image projection. *Nature* 221, 963–964.
- Bach-y-Rita, P., et al., 1969b. A tactile vision substitution system. *Am. J. Opt. Arch. Am. A* 46, 109–111.
- Bach-y-Rita, P., 1979. The practicality of sensory aids. *Int. Rehabil. Med.* 1, 87–89.
- Bach-y-Rita, P., 1983. Tactile vision substitution: past and future. *Int. J. Neurosci.* 19, 29–36.
- Bach-y-Rita, P., et al., 1998. Form perception with a 49-point electrotactile stimulus array on the tongue: a technical note. *J. Rehabil. Res. Dev.* 35, 427–430.
- Bach-y-Rita, P., Kercel, S.W., 2003. Sensory substitution and the human-machine interface. *Trends Cogn. Sci.* 7, 541–546.
- Bach-y-Rita, P., et al., 2003. Seeing with the brain. *Int. J. Hum. Comput. Int.* 15, 285–295.
- Bavelier, D., Neville, H.J., 2002. Cross-modal plasticity: where and how? *Nat. Rev. Neurosci.* 3, 443–452.
- Bhattacharjee, A., et al., 2010. Vibrotactile masking experiments reveal accelerated somatosensory processing in congenitally blind braille readers. *J. Neurosci.* 30, 14288–14298.
- Brown, D., et al., 2011. Seeing with sound? exploring different characteristics of a visual-to-auditory sensory substitution device. *Perception* 40, 1120–1135.
- Büchel, C., et al., 1998. Different activation patterns in the visual cortex of late and congenitally blind subjects. *Brain* 121, 409–419.
- Burton, H., et al., 2002. Adaptive changes in early and late blind: a fMRI study of Braille reading. *J. Neurophysiol.* 87, 589–607.
- Capelle, C., et al., 1998. A real-time experimental prototype for enhancement of vision rehabilitation using auditory substitution. *IEEE Trans. Biomed. Eng.* 45, 1279–1293.
- Chebat, D.R., et al., 2007. Tactile-‘visual’ acuity of the tongue in early blind individuals. *Neuroreport* 18, 1901–1904.
- Chebat, D.R., et al., 2011. Navigation with a sensory substitution device in congenitally blind individuals. *Neuroreport* 22, 342–347.
- Cholewiak, R.W., Craig, J.C., 1984. Vibrotactile pattern recognition and discrimination at several body sites. *Percept. Psychophys.* 35, 503–514.
- Cholewiak, R.W., Collins, A.A., 1997. Individual differences in the vibrotactile perception of a “simple” pattern set. *Percept. Psychophys.* 59, 850–866.
- Cohen, L.G., et al., 1997. Functional relevance of cross-modal plasticity in blind humans. *Nature* 389, 180–183.
- Cohen, L.G., et al., 1999. Period of susceptibility for cross-modal plasticity in the blind. *Ann. Neurol.* 45, 451–460.
- Collignon, O., et al., 2009. Cross-modal plasticity for the spatial processing of sounds in visually deprived subjects. *Exp. Brain Res.* 192, 343–358.
- Craig, J.C., 1977. Vibrotactile pattern perception: extraordinary observers. *Science* 196, 450–452.
- Craig, J.C., 1981. Tactile letter recognition: pattern duration and modes of pattern generation. *Percept. Psychophys.* 30, 540–546.
- Danilov, Y., Tyler, M., 2005. Brainport: an alternative input to the brain. *J. Integr. Neurosci.* 4, 537–550.
- De Volder, A.G., et al., 1999. Changes in occipital cortex activity in early blind humans using a sensory substitution device. *Brain Res.* 826, 128–134.
- Eckhorn, R., et al., 2006. Visual resolution with retinal implants estimated from recordings in cat visual cortex. *Vision Res.* 46, 2675–2690.
- Epstein, W., et al., 1989. Perceptual learning of spatiotemporal events: evidence from an unfamiliar modality. *J. Exp. Psychol. Hum. Percept. Perform.* 15, 28–44.
- Fieger, A., et al., 2006. Auditory spatial tuning in late-onset blindness in humans. *J. Cogn. Neurosci.* 18, 149–157.
- Fine, I., et al., 2003. Long-term deprivation affects visual perception and cortex. *Nat. Neurosci.* 6, 915–916.
- Goldish, L.H., Taylor, H.E., 1974. The Optacon: a valuable device for blind persons. *New Outlook Blind* 68, 49–56.
- Goldreich, D., Kanics, I.M., 2003. Tactile acuity is enhanced in blindness. *J. Neurosci.* 23, 3439–3445.

- Grant, A.C., et al., 2000. Tactile perception in blind braille readers: a psychophysical study of acuity and hyperacuity using gratings and dot patterns. *Percept. Psychophys.* 62, 301–312.
- Hamilton, R., et al., 2000. Alexia for Braille following bilateral occipital stroke in an early blind woman. *Neuroreport* 11, 237–240.
- Heller, M.A., 1989. Texture perception in sighted and blind observers. *Percept. Psychophys.* 45, 49–54.
- Kauffman, T., et al., 2002. Braille character discrimination in blindfolded human subjects. *Neuroreport* 13, 571–574.
- Kim, J.K., Zatorre, R.J., 2008. Generalized learning of visual-to-auditory substitution in sighted individuals. *Brain Res.* 1242, 263–275.
- Kim, J.K., Zatorre, R.J., 2010. Can you hear shapes you touch?. *Exp. Brain Res.* 202, 747–754.
- Kolarik, A.J., et al., 2013. An assessment of virtual auditory distance judgments among blind and sighted listeners. In: *Proceedings of Meetings on Acoustics*, vol. 19, Montreal, Canada, pp. 1–5.
- Kupers, R., et al., 2006. Transcranial magnetic stimulation of the visual cortex induces somatotopically organized qualia in blind subjects. *Proc. Natl. Acad. Sci. U.S.A.* 103, 13256–13260.
- Kupers, R., et al., 2010. Neural correlates of virtual route recognition in congenital blindness. *Proc. Natl. Acad. Sci. U.S.A.* 107, 12716–12721.
- Kupers, R., Ptito, M., 2014. Compensatory plasticity and cross-modal reorganization following early visual deprivation. *Neurosci. Biobehav. Rev.* 41, 36–52.
- Lazzouni, L., Lepore, F., 2014. Compensatory plasticity: time matters. *Front. Hum. Neurosci.* 8.
- Leclerc, C., et al., 2000. Brain functional reorganization in early blind humans revealed by auditory event-related potentials. *Neuroreport* 11, 545–550.
- Lee, V.K., et al., 2014. Successful tactile based visual sensory substitution use functions independently of visual pathway integrity. *Front. Hum. Neurosci.* 8.
- Lenay, C., et al., 2003. Sensory substitution: limits and perspectives. In: Hatwell, Y., Streri, A., Gentaz, E. (Eds.), *Touching for Knowing*. John Benjamins B.V., Amsterdam, pp. 275–292.
- Lessard, N., et al., 1998. Early-blind human subjects localize sound sources better than sighted subjects. *Nature* 395, 278–280.
- Levy-Tzedek, S., et al., 2012. Fast, accurate reaching movements with a visual-to-auditory sensory substitution device. *Restor. Neurol. Neurosci.* 30, 313–323.
- Linville, J.G., Bliss, J.C., 1966. A direct translation reading aid for the blind. *Proc. IEEE* 54, 40–51.
- Liu, Y., et al., 2007. Whole brain functional connectivity in the early blind. *Brain* 130, 2085–2096.
- Maidenbaum, S., et al., 2014. Sensory substitution: closing the gap between basic research and widespread practical visual rehabilitation. *Neurosci. Biobehav. Rev.* 41, 3–15.
- Matteau, I., et al., 2010. Beyond visual, aural and haptic movement perception: hMT+ is activated by electrotactile motion stimulation of the tongue in sighted and in congenitally blind individuals. *Brain Res. Bull.* 82, 264–270.
- Meijer, P.B.L., 1992. An experimental system for auditory image representations. *IEEE Trans. Biomed. Eng.* 39, 112–121.
- Melzer, P., et al., 2001. Cortical activation during Braille reading is influenced by early visual experience in subjects with severe visual disability: a correlational fMRI study. *Hum. Brain Mapp.* 14, 186–195.
- Merzenich, M.M., et al., 1983. Progression of change following median nerve section in the cortical representation of the hand in areas 3b and 1 in adult owl and squirrel monkeys. *Neuroscience* 10 (639–641), 643–665.
- Merzenich, M.M., et al., 1984. Somatosensory cortical map changes following digit amputation in adult monkeys. *J. Comp. Neurol.* 224, 591–605.
- Murphy, M.C., et al., 2014. Top down influence on the visual cortex of the blind during auditory sensory substitution. In: *Proceedings of International Society for Magnetic Resonance in Medicine*, Milan, Italy.
- Murphy, M.C., et al., 2015. Relationship between visual functional connectivity and duration of blindness depends on onset of visual deprivation. In: *Proceedings of International Society for Magnetic Resonance in Medicine*, Toronto, Ontario.
- Nadol, J.B.J., 1997. Patterns of neural degeneration in the human cochlea and auditory nerve: implications for cochlear implantation. *Otolaryngol. Head Neck Surg.* 117, 220–228.
- Nau, A., et al., 2013. Clinical tests of ultra-low vision used to evaluate rudimentary visual perceptions enabled by the BrainPort vision device. *Transl. Vis. Sci. Technol.* 2, 1–12.
- Nau, A.C., et al., 2014. A standardized obstacle course for assessment of visual function in ultra low vision and artificial vision. *J. Vis. Exp.* 84, e51205.
- Nau, A.C., et al., 2015. Acquisition of visual perception in blind adults using the BrainPort artificial vision device. *Am. J. Occup. Ther.* 69 6901290010p1–6901290010p8.
- Noppeney, U., 2007. The effects of visual deprivation on functional and structural organization of the human brain. *Neurosci. Biobehav. Rev.* 31, 1169–1180.
- Occelli, V., et al., 2013. Auditory, tactile, and audiotactile information processing following visual deprivation. *Psychol. Bull.* 139, 189–212.
- Occelli, V., et al., 2014. Loss of form vision impairs spatial imagery. *Front. Hum. Neurosci.* 8.
- Ortiz, T., et al., 2011. Recruitment of occipital cortex during sensory substitution training linked to subjective experience of seeing in people with blindness. *PLoS ONE* 6.
- Pascual-Leone, A., et al., 2005. The plastic human brain cortex. *Annu. Rev. Neurosci.* 28, 377–401.
- Poirier, C., et al., 2006a. Assessment of sensory substitution prosthesis potentialities in minimalist conditions of learning. *Appl. Cogn. Psychol.* 20, 447–460.
- Poirier, C., et al., 2007a. Pattern recognition using a device substituting audition for vision in blindfolded sighted subjects. *Neuropsychologia* 45, 1108–1121.
- Poirier, C., et al., 2007b. What neuroimaging tells us about sensory substitution. *Neurosci. Biobehav. Rev.* 31, 1064–1070.
- Poirier, C.C., et al., 2006b. Neural changes in the ventral and dorsal visual streams during pattern recognition learning. *Neurobiol. Learn. Mem.* 85, 36–43.
- Ptito, M., et al., 2005. Cross-modal plasticity revealed by electrotactile stimulation of the tongue in the congenitally blind. *Brain* 128, 606–614.
- Ptito, M., et al., 2008. TMS of the occipital cortex induces tactile sensations in the fingers of blind Braille readers. *Exp. Brain Res.* 184, 193–200.
- Ptito, M., et al., 2009. Recruitment of the middle temporal area by tactile motion in congenital blindness. *Neuroreport* 20, 543–547.
- Ptito, M., et al., 2012. Crossmodal recruitment of the ventral visual stream in congenital blindness. *Neural Plast.* 2012, 1–10.
- Rauschecker, J.P., 1995. Compensatory plasticity and sensory substitution in the cerebral cortex. *Trends Neurosci.* 18, 36–43.
- Reich, L., et al., 2012. The brain as a flexible task machine: implications for visual rehabilitation using noninvasive vs. invasive approaches. *Curr. Opin. Neurol.* 25, 86–95.
- Renier, L., et al., 2005. Cross-modal activation of visual cortex during depth perception using auditory substitution of vision. *NeuroImage* 26, 573–580.

- Renier, L., de Volder, A.G., 2005. Cognitive and brain mechanisms in sensory substitution of vision: a contribution to the study of human perception. *J. Integr. Neurosci.* 4, 489–503.
- Renier, L., De Volder, A.G., 2010. Vision substitution and depth perception: early blind subjects experience visual perspective through their ears. *Disabil. Rehabil. Assist. Technol.* 5, 175–183.
- Röder, B., et al., 1996. Event-related potentials during auditory and somatosensory discrimination in sighted and blind human subjects. *Cogn. Brain Res.* 4, 77–93.
- Röder, B., et al., 1999. Improved auditory spatial tuning in blind humans. *Nature* 400, 162–166.
- Sadato, N., et al., 1998. Neural networks for Braille reading by the blind. *Brain* 121, 1213–1229.
- Sadato, N., et al., 2002. Critical period for cross-modal plasticity in blind humans: a functional MRI study. *NeuroImage* 16, 389–400.
- Sampaio, E., et al., 2001. Brain plasticity: ‘Visual’ acuity of blind persons via the tongue. *Brain Res.* 908, 204–207.
- Sathian, K., Zangaladze, A., 1997. Tactile learning is task specific but transfers between fingers. *Percept. Psychophys.* 59, 119–128.
- Sathian, K., 2000. Practice makes perfect: sharper tactile perception in the blind. *Neurology* 54, 2203–2204.
- Sathian, K., Stilla, R., 2010. Cross-modal plasticity of tactile perception in blindness. *Restor. Neurol. Neurosci.* 28, 271–281.
- Segond, H., et al., 2005. Human spatial navigation via a visuo-tactile sensory substitution system. *Perception* 34, 1231–1249.
- Segond, H., et al., 2013. Perceiving space and optical cues via a visuo-tactile sensory substitution system: a methodological approach for training of blind subjects for navigation. *Perception* 42, 508–528.
- Sharma, A., et al., 2002. A sensitive period for the development of the central auditory system in children with cochlear implants: implications for age of implantation. *Ear Hear.* 23, 532–539.
- Shimojo, S., Shams, L., 2001. Sensory modalities are not separate modalities: plasticity and interactions. *Curr. Opin. Neurobiol.* 11, 505–509.
- Simon, H.J., et al., 2002. Lateralization of narrow-band noise by blind and sighted listeners. *Perception* 31, 855–873.
- Spengler, F., et al., 1997. Learning transfer and neuronal plasticity in humans trained in tactile discrimination. *Neurosci. Lett.* 232, 151–154.
- Sterr, A., et al., 1998. Perceptual correlates of changes in cortical representation of fingers in blind multifinger Braille readers. *J. Neurosci.* 18, 4417–4423.
- Striem-Amit, E., et al., 2012. ‘Visual’ acuity of the congenitally blind using visual-to-auditory sensory substitution. *PLoS ONE* 7.
- Stronks, H.C., Dagnelie, G., 2014. The functional performance of the Argus II retinal prosthesis. *Expert Rev. Med. Devices* 11, 23–30.
- Théoret, H., et al., 2004. Behavioral and neuroplastic changes in the blind: evidence for functionally relevant cross-modal interactions. *J. Physiol.—Paris* 98, 221–233.
- Thinus-Blanc, C., Gaunet, F., 1997. Representation of space in blind persons: vision as a spatial sense?. *Psychol. Bull.* 121, 20–42.
- Van Boven, R.W., et al., 2000. Tactile spatial resolution in blind Braille readers. *Neurology* 54, 2230–2236.
- Veraart, C., et al., 1990. Glucose utilization in human visual cortex is abnormally elevated in blindness of early onset but decreased in blindness of late onset. *Brain Res.* 510, 115–121.
- Wan, C.Y., et al., 2010. Congenital blindness leads to enhanced vibrotactile perception. *Neuropsychologia* 48, 631–635.
- Wong, M., et al., 2011. Tactile spatial acuity enhancement in blindness: evidence for experience-dependent mechanisms. *J. Neurosci.* 31, 7028–7037.